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Research Brief

Maternal ADHD Diagnoses Before and After Childbirth: A Danish Population-Based Cohort Study

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Abstract

Objective: The rates of women of reproductive age diagnosed with ADHD have increased significantly. However, little is known about how pregnancy, childbirth, and the postpartum period influence ADHD related problems, help-seeking behaviors or the timing of ADHD diagnosis. This study aimed to investigate incident ADHD diagnoses among women from 5 years before to 5 years after childbirth. **Method:** The population-based cohort study utilized Danish nationwide registers to identify mothers of all childbirths between 2010 and 2018. ADHD diagnoses, defined as ICD-10 codes (F90 or F98.8) or redemption of ADHD medication prescriptions, were recorded during 1994 to 2023. Women who received an ADHD diagnosis prior to the five-year pre-birth period were excluded. We estimated calendar-year adjusted incidence of first-time maternal ADHD diagnoses using Poisson regression, and compared incidence across pre-pregnancy, pregnancy, postpartum up to 2 years, and 2 to 5 years postpartum, reporting adjusted incidence rate ratios (IRRs). Analyses included psychiatric comorbidities and non-ADHD psychotropic medication use prior to an ADHD diagnosis, for women diagnosed with ADHD postpartum. **Results:** The study included 363,904 mothers with a total of 524,936 childbirths. ADHD incidence rates decreased significantly during pregnancy (IRR = 0.28, 95% CI [0.22, 0.37]), remained lower than the pre-pregnancy period 2 years postpartum (IRR 0.82, 95% CI [0.74, 0.92]), and increased significantly above pre-pregnancy levels 2 to 5 years postpartum (IRR = 1.24, 95% CI [1.13, 1.35]). Among mothers diagnosed with ADHD postpartum, 53.9% had an in- or outpatient contact to psychiatric services or redeemed non-ADHD psychotropic medication prescriptions, in the period from childbirth until ADHD diagnosis. **Conclusion:** Incident ADHD diagnoses among women giving birth increased postpartum above pre-pregnancy levels, with peak rates observed between 2 and 5 years after childbirth. These findings highlight the postpartum period as a potential critical window of ADHD symptom worsening, underscoring the need for targeted mental health screenings and support for women during the years after childbirth. (*J. of Att. Dis. 2026; 30(3) 283-291*)

Keywords

ADHD, pregnancy, childbirth, postpartum psychiatric disorders, incidence

Introduction

ADHD diagnoses have increased significantly among women of reproductive age in recent years, possibly driven by heightened awareness, recognition of the disorder's persistence into adulthood, improved diagnostic practices, and evolving societal perceptions (Anderson et al., 2018; Kooij et al., 2019; Scoten et al., 2024; Zetterqvist et al., 2013). Many women with ADHD, however, experience delayed diagnosis (Skoglund et al., 2024), often due to difficulties with recognizing ADHD symptoms in girls and women, which are frequently masked by compensatory behaviors or misinterpreted as mood or anxiety disorders (Attoe & Climie, 2023; Madsen et al., 2018). The increased attention to females with ADHD and their common diagnostic delay emphasize the existing gap in understanding how life events unique to this population, such as childbirth and the perinatal period, affect symptomatology, help-seeking behaviors and the timing of ADHD diagnosis.

Giving birth represents a profound physiological and psychological transition. Cognitive impairment and decreased executive functioning, key symptoms of ADHD, are more common during pregnancy than in non-pregnant states (Davies et al., 2018). Hormonal fluctuations during pregnancy and the postpartum period may contribute to neurobiological alterations that exacerbate ADHD symptoms,

including changes in brain structure and function (Hoekzema et al., 2017; Rehbein et al., 2021). These changes, combined with the heightened demands of pregnancy and early parenthood, may amplify distractibility and cognitive fatigue, posing additional challenges for women with ADHD. Eddy et al. (2019) found that inattentive symptoms were significant predictors of professional life impairment, daily life impairment, and relationship impairment, while impulsivity uniquely predicted variability in professional life and relationship impairment in pregnant women with ADHD. The postpartum period may further exacerbate these challenges. New or worsened cognitive impairments, disruptions to previously effective daily strategies, and the demands of caring for a newborn may increase ADHD symptoms and functional impairments, possibly prompting help-seeking behaviors. Evidence suggests that parents with ADHD and their partners experience greater parenting distress in the first year of their child's life than parents without ADHD (Joseph et al., 2022). Also, postpartum depression, a common condition for mothers with ADHD (Andersson et al., 2023), may also act as a trigger for ADHD diagnosis, as its overlapping symptoms, such as emotional dysregulation and executive dysfunction, may prompt clinical attention to previously unrecognized ADHD.

This study investigates whether the transition into parenthood, including pregnancy, childbirth, potential postpartum psychiatric comorbidity, and the demands of caring for a young child, drives maternal help-seeking, as reflected by the timing of first-time ADHD diagnoses. We hypothesize that this transition will amplify previously unrecognized ADHD symptoms and thereby prompt women to seek evaluation and care. Consequently, we expect a rise in incident first-time maternal ADHD diagnoses during the postpartum years exceeding pre-pregnancy levels.

Methods

Study Design and Population

This descriptive population-based cohort study utilized data from Danish nationwide registers. All childbirths occurring from 2010 to 2018 were identified through the Danish Medical Birth Registry (Bliddal et al., 2018). After excluding 8,991 childbirths due to missing or unlikely gestational ages (<154 or >315 days) and 3,326 childbirths by mothers with a preexisting record of ADHD diagnosis before the 5-year pre-birth period, the final cohort included 363,904 mothers to 524,936 liveborn children.

Incidence of ADHD Before and After Childbirth

First-time records of ADHD were considered each month, starting at 5 years before until 5 years after childbirth, with 1 month defined as 30.44 days. To ensure only incident diagnoses were included, we excluded women with an ADHD diagnosis recorded between 1994 and 2005 or before the 5-year pre-birth period leaving enough time to exclude women who had received a diagnosis in childhood or adolescence. The 5-year pre-childbirth window provides sufficient time to establish a stable baseline for ADHD incidence, while extending follow-up to 5 years postpartum accounts for the possibility of delayed diagnoses between initial symptom recognition and formal diagnostic recording. An incident diagnosis of ADHD was defined as an ICD-10 F90 or F98.8 diagnosis identified in the Danish National Patient Register (Lynge et al., 2011) or a first time use of ADHD medication using the following Anatomical Therapeutic Chemical (ATC) classification codes for stimulant treatment: N06BA04 "Methylphenidate," N06BA01 "Amphetamine," N06BA02 "Dexamfetamine," N06BA12 "Lisdexamfetamine," and other treatment: N06BA07 "Modafinil," and N06BA09 "Atomoxetine" in the Danish National Prescription Register (Kildemoes et al., 2011). In Denmark, citizens have the right to consult privately practicing specialists free of charge if waiting times at public hospital services exceed 1 month. However, private psychiatrists are not required to report diagnoses to national registries. Therefore, we used redeemed ADHD medication prescriptions as a proxy for ADHD diagnoses, since only psychiatrists are authorized to prescribe these medications following a comprehensive clinical assessment confirming ADHD. Recorded ADHD diagnoses in the Danish registers are deemed valid and reliable (Mohr-Jensen et al., 2016). Among postpartum ADHD diagnosed mothers, we estimated the proportion with in- and outpatient admissions

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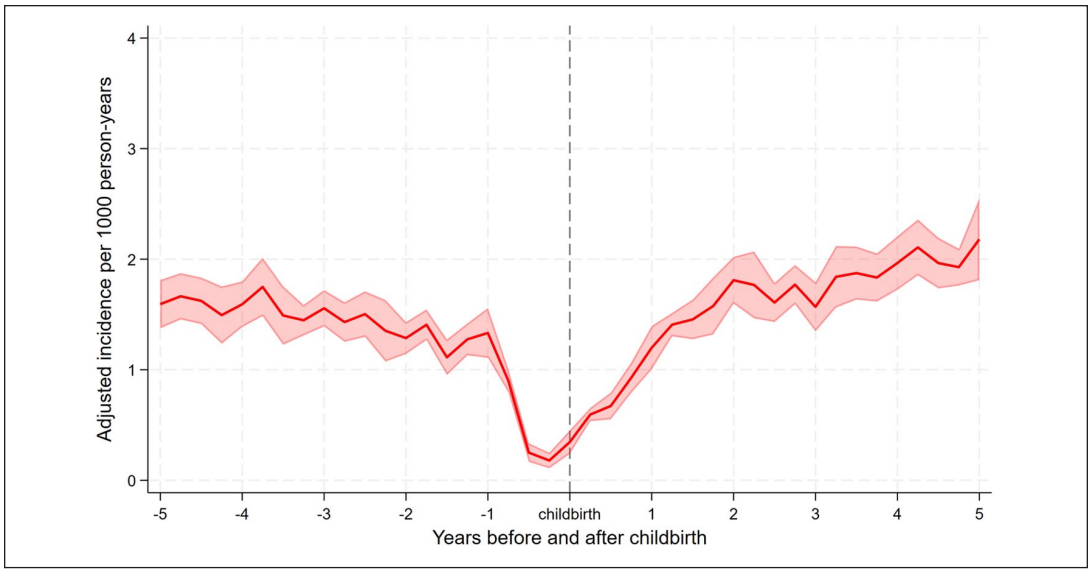


Figure 1. Calendar year adjusted ADHD incidence rates 5 years before to 5 years after childbirth per 1,000 person-years—all childbirths during 2010 to 2018.

with the ICD-10 diagnostic codes F10 to F19 "Mental and behavioral disorders due to psychoactive substance use," F30 to F39 "Mood disorders," and F40 to F48 "Neurotic, stress-related, and somatoform disorders" and non-ADHD psychotropic medication prescription fills; antidepressants (ATC N06A), anxiolytics (ATC N05B), and drugs used in alcohol dependence (ATC N07BB) in the period from childbirth until ADHD diagnosis.

Statistical Analyses

Women were followed from 5 years pre-childbirth until an ADHD diagnosis, 5 years after childbirth, or December 2023, whichever occurred first. The incidence was based on the first-ever ADHD diagnosis within the period from 5 years before through 5 years after birth; meaning that incident cases could not have had earlier records of ADHD treatment. To estimate the incidence of first-time maternal ADHD diagnosis across different time periods surrounding childbirth adjusting for calendar year of diagnosis, we fitted Poisson regression models with robust standard errors. The dependent variable was the count of ADHD diagnoses in each quarter from 5 years before to 5 years after childbirth. The offset term was the natural logarithm of person-years observed in each quarter, allowing us to model incidence rates with 95% CI. We defined four exposure periods relative to childbirth: pre-pregnancy (5 years to 9 months before childbirth), pregnancy (9 months before to childbirth), postpartum 0 to 2 years, and postpartum 2 to 5 years to compare incidence rates in these periods. The Poisson model included the period variable as a categorical predictor and adjusted for calendar year to account for secular trends in ADHD diagnosis. Adjusted incidence rate ratios (IRRs) and 95% confidence intervals (CIs) were estimated using the IRR option in Stata. Marginal incidence rates (per 1,000 person-years) for each period were estimated using the margins command. Also, we determined the proportion diagnosed with other psychiatric disorders or filling prescriptions for non-ADHD psychotropics among mothers who received an ADHD diagnosis in the 5-year postpartum period. We restricted this comorbidity analysis to the interval between childbirth and the mother's first ADHD diagnosis postpartum, because this window provides a uniform, diagnosis-anchored frame, whereas an equivalent pre-pregnancy interval would be arbitrary and capture temporally distant disorders that are not directly comparable to the time period between childbirth and initial ADHD diagnosis. Finally, we conducted a sensitivity analysis including a random sample of one childbirth per mother to take into account the possible counting of the same mother multiple times across different childbirths.

Analyses were conducted using Stata (version 18 StataCorp, College Station, TX, USA).

Results

Between 2010 and 2018, our study included 363,904 mothers who gave birth to 524,936 children. During the 10-year period surrounding childbirth (5 years before and 5 years after), 7,940 childbirths were associated with a maternal ADHD diagnosis, corresponding to 7,077 unique mothers. Of these, 2,974 mothers received their diagnosis within the 5 years before childbirth, and 4,103 within the 5 years after childbirth. Maternal mean age at first childbirth was significantly lower for

Table 1. Calendar Year Adjusted ADHD Incidence Rates in the First Quarter From 5 Years Before to 5 Years After Childbirth per 1,000 Person-years—All Childbirths During 2010 to 2018.

	All childbirths ($N = 524,936$)	
Time in relation to childbirth	Cases (n)	Adjusted incidence rate per 1,000 person, years [95% CI]
−5 years (Q1)	172	1.59 [1.35, 1.83]
−4 years (Q1)	192	1.59 [1.37, 1.82]
−3 years (Q1)	202	1.56 [1.34, 1.77]
−2 years (Q1)	175	1.29 [1.10, 1.48]
−1 years (Q1)	187	1.33 [1.14, 1.52]
Childbirth (Q1)	48	0.35 [0.25, 0.44]
+1 years (Q1)	162	1.19 [1.01, 1.38]
+2 years (Q1)	244	1.81 [1.58, 2.04]
+3 years (Q1)	223	1.57 [1.36, 1.77]
+4 years (Q1)	321	1.97 [1.75, 2.18]
+5 years	141	2.18 [1.82, 2.57]

Table 2. Test for Period Difference With Poisson Regression Adjusted for Calendar Year of Diagnosis.

	All childbirths ($N = 524,936$)		
Period in relation to childbirth	IRR	95% CI	p
Pre-pregnancy	ref	ref	—
Pregnancy	0.28	[0.22, 0.37]	>.001
Postpartum 0–2 years	0.82	[0.74, 0.92]	>.001
Postpartum 2-5 years	1.24	[1.13, 1.35]	>.001

Note. IRR = incidence rate ratio; 95 CI = 95% confidence interval.

Table 3. In- and Outpatient Psychiatric Contact and/or Filling of Psychotropic Medication From Birth Until 5 Years Postpartum Prior to ADHD Diagnosis.

	ADHD diagnosis postpartum ($N = 4,597$)	
Psychiatric contact and/or psychotropic medication fills	Childbirths, n	%
Contact and/or medication	2,476	53.9
Contact all	842	18.3
Substance use disorder	57	1.2
Mood disorder	448	9.8
Anxiety disorder	529	11.5
Medication all	2,293	49.9
Drugs for alcohol dependence	46	1.0
Antidepressants	2,181	47.4
Anxiolytics	411	8.9

mothers who received an ADHD diagnosis in the period from 5 years pre-birth to 5 years postpartum (mean age 26.9 years [$SD = 5.3$]) compared to mothers who did not (mean age 29.3 years [$SD = 4.8$]; t-test, $p < .001$).

Calendar year–adjusted incidence rates with 95% CI of first-time maternal ADHD diagnosis are presented quarterly across the study period in Figure 1 and for the first quarter each year in Table 1. The adjusted incidence rate 5 years before childbirth was 1.59 per 1,000 person-years (95% CI [1.35, 1.83]) and increased to 2.18 (95% CI [1.82, 2.57]) 5 years after childbirth, which can be interpreted as one new ADHD diagnosis for every 630 and 460 women, respectively. As presented in Table 2, compared with the pre-pregnancy period (defined as 5 years to 9 months before childbirth), the incidence was significantly lower during pregnancy (IRR = 0.28, 95% CI [0.22, 0.37]). In the first 2 years postpartum, the adjusted incidence was slightly but significantly lower compared to the pre-pregnancy period (IRR = 0.82, 95% CI [0.74, 0.92]). A significant increase was observed in the period from 2 to 5 years postpartum, peaking at 4 to 5 years postpartum, where the adjusted incidence was significantly higher than in the pre-pregnancy period (IRR = 1.24, 95% CI [1.13, 1.35]).

Of the 4,597 childbirths where mothers received an ADHD diagnosis within 5 years postpartum, 2,476 (53.9%) involved mothers who were diagnosed with depression, anxiety, or substance use disorder or redeemed prescriptions for antidepressants, anxiolytics, or alcohol dependence medication before their initial ADHD diagnosis (Table 3). The same pattern was observed in the sensitivity analysis restricting to one random childbirth per mother (Figure 2, Tables 4–6).

Discussion

This study demonstrated that ADHD diagnosis incidence decreased during pregnancy but increased significantly in the postpartum period, reaching pre-pregnancy levels between 1 and 2 years postpartum and continuing to rise, peaking at about 4 years postpartum. Additionally, a majority of mothers were diagnosed with depression, anxiety, or substance use disorder or redeemed prescriptions for non-ADHD psychotropics in the period from childbirth until receiving their ADHD diagnosis postpartum.

The significant decline in ADHD incidence during pregnancy may reflect multiple factors, including symptom reduction or masking due to hormonal changes (Eng et al., 2024), lower likelihood of help-seeking during this period, or hesitancy among healthcare providers to diagnose ADHD or initiate treatment during pregnancy (Hinshaw et al., 2022). This may lead to unmet needs that resurface postpartum where ADHD incidence increased above pre-birth levels.

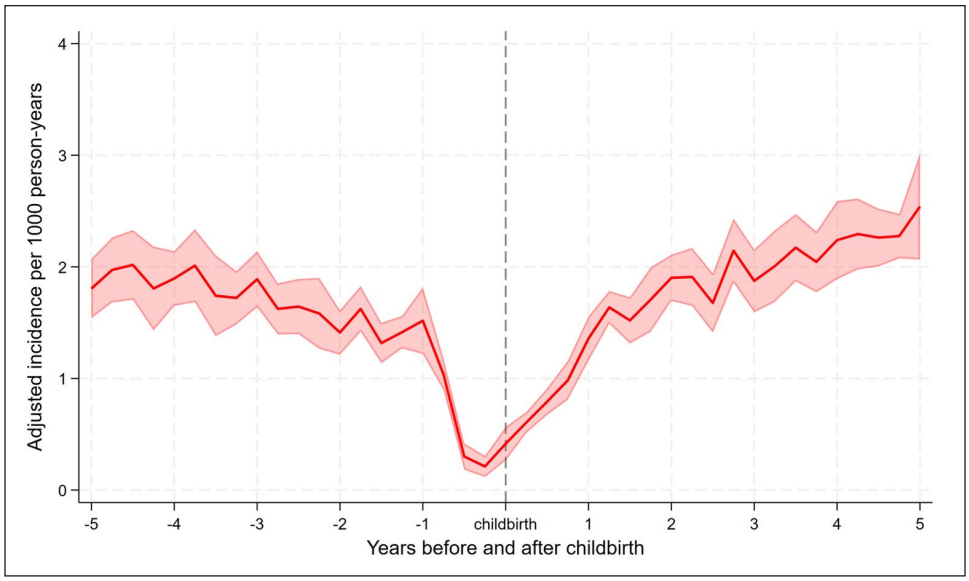


Figure 2. Sensitivity analysis: Calendar year adjusted ADHD incidence rates 5 years before to 5 years after childbirth per 1,000 person-years—one randomly selected childbirth per mother.

Table 4. Calendar Year Adjusted ADHD Incidence Rates in the First Quarter From 5 Years Before to 5 Years After Childbirth per 1,000 person-years—one random childbirth during 2010 to 2018.

	All childbirths ($N = 524{,}936$)	
Time in relation to childbirth	Cases (*n*)	Adjusted incidence rate per 1,000 person, years [95% CI]
−5 years (Q1)	142	1.79 [1.50, 2.08]
−4 years (Q1)	167	1.91 [1.63, 2.21]
−3 years (Q1)	273	1.85 [1.58, 2.13]
−2 years (Q1)	135	1.38 [1.15, 1.61]
−1 years (Q1)	161	1.59 [1.35, 1.84]
Childbirth (Q1)	43	0.43 [0.30, 0.57]
+1 years (Q1)	175	1.93 [1.65, 2.22]
+2 years (Q1)	161	1.75 [1.48, 2.03]
+3 years (Q1)	246	2.36 [2.07, 2.65]
+4 years (Q1)	94	2.29 [1.83, 2.76]
+5 years	141	2.18 [1.82, 2.57]

Table 5. Sensitivity Analysis: Test for Period Difference With Poisson Regression Adjusted for Calendar Year of Diagnosis.

	One random childbirth per mother ($N = 363{,}904$)		
Period in relation to childbirth	IRR	95% CI	*p*
Pre-pregnancy	ref	ref	—
Pregnancy	0.29	[0.23, 0.38]	>.001
Postpartum 0–2 years	0.76	[0.68, 0.86]	>.001
Postpartum 2–5 years	1.21	[1.10, 1.33]	>.001

Note. IRR = incidence rate ratio; 95 CI = 95% confidence interval.

Also, the challenges associated with parenting, combined with the cumulative cognitive and emotional demands of the postpartum period, may highlight previously unrecognized ADHD symptoms. The caregiving task itself, usually including deprived sleep and emotional outbursts or demands of the infant or preschooler, might amplify ADHD core symptoms, and impairment may initially present as depression or anxiety, which, in our data, often preceded the ADHD diagnosis (Joseph et al., 2022).

We observed a peak in ADHD incidence 4 to 5 years after childbirth, which could be due to the documented delay between women's first recognition of ADHD-related difficulties and their action upon them (Agnew-Blais, 2024; Skoglund et al., 2024). Women with ADHD, and their relatives, may initially attribute inattentiveness, disorganization, and emotional lability to regular postpartum exhaustion or parenting stress. It is possible that formal evaluation is not requested until these problems have persisted for years.

Strikingly more than half of mothers diagnosed with ADHD postpartum had other psychopathology (defined as in- or outpatient contacts for depression, anxiety, or substance use disorders, or redeemed prescriptions for non-ADHD psychotropic medications) before receiving their ADHD diagnosis. These comorbidities often mask ADHD symptoms and complicate the assessment of ADHD, which

Table 6. Sensitivity Analysis—In- and Outpatient Psychiatric Contact and/or Filling of Psychotropic Medication From Birth Until 5 Years Postpartum Prior to ADHD Diagnosis.

	ADHD diagnosis postpartum ($N = 3,384$)	
Psychiatric contact and/or psychotropic medication fills	Childbirths, *n*	%
Contact and/or medication	1,837	54.3
Contact all	645	19.1
Substance use disorder	43	1.3
Mood disorder	343	10.1
Anxiety disorder	403	11.9
Medication all	1,697	50.2
Drugs for alcohol dependence	33	1.0
Antidepressants	1,619	47.8
Anxiolytics	300	8.9

may cause physicians to overlook the diagnosis of ADHD (Young et al., 2020) contributing to diagnostic delays or even misdiagnosis (Quinn & Madhoo, 2014). Conversely, postpartum depression symptoms, such as concentration difficulties (inattention) and psychomotor agitation (hyperactivity/impulsivity), may exaggerate ADHD symptoms, increasing the risk of temporarily pushing those with elevated but still normative levels of ADHD symptoms into the clinical range.

We found a significantly lower maternal mean age at first childbirth for mothers with ADHD compared to mothers without in line with previous research (Amikam et al., 2024; Skoglund et al., 2019). A recent study found a four-times higher rate of unplanned pregnancies in young women with ADHD compared to non-ADHD controls (Owens & Hinshaw, 2020). Such unintended, and often early pregnancies, are consistently associated with adverse obstetric and perinatal outcomes (Skoglund et al., 2019) that are also linked to postpartum psychiatric disorders (Meltzer-Brody et al., 2017).

Our findings emphasize the need for heightened clinical awareness of ADHD in women during the perinatal period. Although limited research has examined the consequences of delayed ADHD diagnosis in women, existing evidence highlights its profound and far-reaching effects. Late diagnosis is associated with diminished self-esteem, significant mental health challenges, disruptions in personal identity, and reduced life opportunities (Attoe & Climie, 2023; Skoglund et al., 2024; Young et al., 2020). These impacts often lead to increased psychosocial burdens and adverse outcomes, including strained relationships and impaired caregiving capacity (Morgan, 2024). To address this, perinatal care providers should be trained to recognize the overlapping symptoms of ADHD, depression, and anxiety, and always consider ADHD as differential diagnosis in women presenting with executive dysfunction or emotional dysregulation. Integrating mental health screenings into routine perinatal care could facilitate earlier identification and treatment, reducing the long-term burden of undiagnosed ADHD for both mother and child.

Strengths and Limitations

The use of a nationwide cohort with comprehensive data on ADHD diagnoses, medication prescriptions, and psychiatric comorbidities is a major strength of this study, allowing us to capture population-wide trends and minimize selection bias. Additionally, the longitudinal design enabled us to examine diagnostic patterns over a 10-year window, providing a nuanced understanding of ADHD incidence relative to childbirth.

However, limitations should be acknowledged. First, ADHD diagnoses were based on clinical records and medication prescriptions, which may underestimate true incidence, particularly for women with mild symptoms, and the absence of data on subjective experiences or self-reported symptoms limits our ability to fully understand the psychosocial factors driving help-seeking behaviors. Second, we deliberately capped follow-up at 5 years postpartum to minimize referral bias from mothers who seek evaluation only after their child is assessed (Chatwin et al., 2025). This restriction, however, prevents us from determining whether the rising diagnostic trend after childbirth plateaus, continues, or declines into child school age. Third, we interpreted a first-time ADHD diagnosis in the registers as a proxy for maternal help-seeking; however, because registry data lack information on the motivations and pathways that lead women to initiate evaluation, this proxy cannot fully capture the complexity of help-seeking behavior.

Future research should move beyond describing the postpartum rise in diagnoses to clarifying the mechanisms that drive it. One plausible pathway is that the heightened executive and emotional demands of early parenthood exacerbate previously unrecognized ADHD symptoms, prompting women to seek help and, ultimately, receive a diagnosis, but this hypothesis remains untested. Likewise, although the striking rates of postpartum depression and anxiety in newly diagnosed mothers raise the possibility that mood symptoms are, in part, downstream consequences of undetected ADHD, the temporal and causal links are still unknown.

Conclusion

This study highlights distinct patterns in ADHD diagnosis incidence among women before and after childbirth. While ADHD diagnoses decreased significantly during pregnancy, incidence rates returned to pre-pregnancy levels by 2 years postpartum and peaked at 4 to 5 years postpartum. The cumulative incidence of ADHD diagnoses was notably

higher in the postpartum period, suggesting that early parenthood is particularly challenging to women with ADHD, which possibly triggers help-seeking eventually leading to diagnostic recognition.

A significant proportion of mothers diagnosed with ADHD postpartum had prior contacts to hospital care for depression, anxiety, or substance use disorders or redeemed non-ADHD psychotropic medications before receiving their ADHD diagnosis. This highlights both the overlap of psychiatric symptoms and the possibility that unrecognized ADHD may contribute to the development of these comorbidities, potentially delaying accurate ADHD diagnosis. These findings emphasize the postpartum period as a critical window for ADHD detection, highlighting the importance of integrating mental health screenings into routine perinatal care to facilitate timely diagnosis and intervention. Early identification of ADHD in mothers could improve outcomes not only for the women themselves but also for their children, reducing the intergenerational burden of untreated ADHD.

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Ethical Considerations

The study was approved by the Danish Data Protection Agency and Statistics Denmark's Scientific Board. Approval from the Ethics Committee was not required. All data has been de-identified and is not recognizable at an individual level.

Consent to Participate

Informed consent is not required for register-based studies according to Danish law.

Author Contributions

KBM and SKS designed the study. KBM conducted the analyses and wrote the first draft of the manuscript. All authors read and approved the final manuscript and agree to be accountable for the work.

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Declaration of Conflicting Interests

The authors declared the following potential conflicts of interest with respect to the research, authorship, and/or publication of this article: KBM has received speakers fee from Medice Nordic within the past 3 years. TMO has received speakers fee from Lundbeck with in the past 3 years. SKS has received speaker's honoraria from Takeda, Medice Arzneitmittel Pütter GmbH & Co KG as well as Janssen in the past 3 years.

Data Availability Statement

Access to individual-level data from Denmark is governed by Danish authorities. These include the Danish Data Protection Agency, the Danish Health Data Authority, the Ethical Committee, and Statistics Denmark. Each scientific project must be approved before initiation, and approval is granted to a specific Danish research institution. Researchers at Danish research institutions may obtain the relevant approval to access data. International researchers may gain data access if governed by a Danish research institution having approval and data access. Code can be made available upon reasonable request.

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